

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def analyze_recursive_iterative(n):

    def recursive_factorial(x):
        if x == 0:
            return 1
        return x * recursive_factorial(x-1)

    def iterative_factorial(x):
        result = 1
        for i in range(1,x+1):
            result *= i
        return result

    fib_calls = 0

    def recursive_fibonacci(x):
        nonlocal fib_calls
        fib_calls += 1

        if x == 0:
            return 0
        if x == 1:
            return 1
        return recursive_fibonacci(x-1) + recursive_fibonacci(x-2)

    def iterative_fibonacci(x):
        if x == 0:
            return 0

        a = 0
        b = 1

        for i in range(x):
            a,b = b, a+b

        return a

    recursive_fact = recursive_factorial(n)
    iterative_fact = iterative_factorial(n)

    recursive_fib = recursive_fibonacci(n)
    iterative_fib = iterative_fibonacci(n)

    recursive_fact_count = n + 1
    iterative_fact_count = n
    recursive_fib_count = fib_calls
    iterative_fib_count = n

    return[
        "Computation Analysis Report",
        "Recursive Factorial: " + str(recursive_fact),
        "Iterative Factorial: " + str(iterative_fact),
        "Recursive Fibonacci: " + str(recursive_fib),
        "Iterative Fibonacci: " + str(iterative_fib),
        "Operation Count Comparison",
        "Recursive Factorial Count: " + str(recursive_fact_count),
        "Iterative Factorial Count: " + str(iterative_fact_count),
        "Recursive Fibonacci Count: " + str(recursive_fib_count),
        "Iterative Fibonacci Count: " + str(iterative_fib_count)
    ]
```